DOMAIN EXPECTATIONS

DevSecOps

Led by **Sushobhit**

Understand what keeps software running, connected, and secure.

Topics progress from foundational concepts to advanced orchestration and exploitation. Newer students should focus on building a strong conceptual foundation. Second-year students are expected to focus on practical implementation and understanding how systems work under the hood.

Shared basics

- What exactly happens when two devices communicate across a network?

Linux

- Commands, the filesystem, permissions, and package management
- Shell scripting, process management, environment variables, and system logs

Networking

- OSI and TCP/IP models, IP addressing, DNS, and DHCP
- HTTP, ports, protocols, firewalls, and NAT
- Network tools, VPNs, and proxies

Git and GitHub

- Commits, branching, merging, pull requests, and .gitignore

Basic scripting

- Automating tasks using Bash or Python

DevOps — Infrastructure & Cloud

- Build practical experience with infrastructure and cloud deployments.

Systems and remote access

- Background services via systemd and daemon management
- SSH and key-based authentication

Web servers and traffic

- Reverse proxies versus load balancers
- Nginx or Apache basics and SSL/TLS certificates

Containers and deployment

- Virtualization versus containerization
- Docker fundamentals: images, writing a Dockerfile, port mapping, volumes, and Docker Compose
- Deploying applications to a PaaS such as Vercel or Render, or to a Linux cloud VM

Cloud IAM and policies

- Managing role-based access control (RBAC)
- Configuring AWS/Azure identity and resource policies, including S3 bucket policies, EC2 roles, and permissions boundaries

CI/CD and automation

- Automating builds, tests, and deployments using GitHub Actions or GitLab CI

Cybersecurity — Offensive / Red Team

- We care more about your approach to understanding systems and the process you follow to exploit them.

Security fundamentals

- The CIA triad and authentication versus authorization
- Understanding cookies, sessions, and JWTs

Cryptography basics

- Encoding (Base64) versus hashing (MD5, SHA-256, bcrypt) versus encryption (symmetric and asymmetric)

Reconnaissance and scanning

- Information gathering, footprinting, and port scanning with Nmap
- Directory fuzzing with ffuf or Gobuster
- Packet inspection with Wireshark or tcpdump

Web exploitation

- Intercepting and modifying HTTP requests with Burp Suite or Caido
- Vulnerabilities including SQLi, XSS, CSRF, IDOR, SSRF, and more
- The OWASP Top 10 is key

System exploitation

- Linux privilege escalation basics: SUID binaries, sudo misconfigurations, and cron jobs
- Understanding reverse and bind shells

Cybersecurity — Defensive / Blue Team

- Secure, monitor, and investigate systems once they are running.

Security Operations (SOC) & monitoring

- Centralized log management and configuring SIEM/SOAR
- Identifying brute-force attempts and detecting anomalous traffic
- Hunting for indicators of compromise (IoCs) in server and network logs

Incident Response (IR)

- The IR lifecycle: Preparation, Identification, Containment, Eradication, and Recovery
- Triaging security alerts to distinguish false positives from actual breaches

Digital Forensics (DFIR)

- Disk and memory imaging concepts and timeline analysis
- Preserving the chain of custody
- Basic malware static analysis: extracting strings and inspecting file signatures and magic bytes

Don't let the heavy terminology panic you. We'll secure the system once it's turned on.

Submitting an application does not confirm selection. Shortlisting and an offline round follow, and their date, time, and venue will be announced.